

Word Sense Disambiguity

AIM : To Create a Sample list for at least 5 words with ambiguous sense and Write a Python program to implement WSD

Description : Word Sense Disambiguation (WSD) is the process of identifying the correct meaning (sense) of a word based on its context, especially when the word has multiple meanings.

Lesk's algorithm is a knowledge-based method for WSD that disambiguates a word by comparing the dictionary definitions (glosses) of its possible senses with the context in which the word appears.

The correct sense of an ambiguous word is the one whose definition overlaps the most with the definitions of the surrounding words in the sentence.

Working of algorithm:

- Identify all possible senses of the ambiguous word using a dictionary like WordNet.
- For each sense, take its definition (gloss).
- Compare it with the glosses or words in the context (surrounding words).
- Count overlapping words between glosses and context.
- Choose the sense with the maximum overlap.

Program:

```
import nltk
from nltk.wsd import lesk
from nltk.corpus import wordnet
# Download required data (run once)
nltk.download('wordnet')
nltk.download('omw-1.4')
# Function for WSD
def get_sense(sentence, ambiguous_word):
    words = sentence.split()
    sense = lesk(words, ambiguous_word)
    return sense
# Test sentences
sentences = [
    ("He deposited money in the bank", "bank"),
    ("The boat reached the bank of the river", "bank"),
    ("The bat flew in the night", "bat"),
    ("He hit the ball with a bat", "bat"),
    ("The crane is flying in the sky", "crane"),
    ("The crane lifted heavy loads", "crane")
]
# Run WSD
for sent, word in sentences:
    sense = get_sense(sent, word)
    print(f"Sentence: {sent}")
    print(f"Word: {word}")
    print(f"Predicted Sense: {sense}")
```

```
print(f"Definition: {sense.definition() if sense else 'No sense found'}")
print("-" * 50)
```

OUTPUT:

```
Sentence: He deposited money in the bank
Word: bank
Predicted Sense: Synset('savings_bank.n.02')
Definition: a container (usually with a slot in the top) for keeping money at home
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Sentence: The boat reached the bank of the river
Word: bank
Predicted Sense: Synset('bank.v.07')
Definition: cover with ashes so to control the rate of burning
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Sentence: The bat flew in the night
Word: bat
Predicted Sense: Synset('cricket_bat.n.01')
Definition: the club used in playing cricket
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Sentence: He hit the ball with a bat
Word: bat
Predicted Sense: Synset('bat.v.01')
Definition: strike with, or as if with a baseball bat
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Sentence: The crane is flying in the sky
Word: crane
Predicted Sense: Synset('grus.n.01')
Definition: a small constellation in the southern hemisphere near Phoenix
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Sentence: The crane lifted heavy loads
Word: crane
Predicted Sense: Synset('crane.n.04')
Definition: lifts and moves heavy objects; lifting tackle is suspended from a pivoted
boom that rotates around a vertical axis
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